

## FORMATION MECHANISM FOR BROAD AND SHALLOW PROFILES OF CYCLOTRON LINES IN ACCRETING X-RAY PULSARS

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### ABSTRACT

We model cyclotron lines by calculating a superposition of the spectra of some cyclotron lines emerging from some individual line-forming regions with a uniform density, temperature, and magnetic field strength in each region. A whole line-forming region is composed of a series of a large number of individual different domains at the temperature, density, and magnetic field strength, which vary with height or location of each of the individual domains. We study the properties of the cyclotron lines formed by superposing some cyclotron lines emerging from different heights or locations of an accretion column or mound and present the influences of the superposition of the lines, a geometry of the line-forming region, and the variations in a density, temperature, and magnetic field on the shapes of cyclotron resonant scattering lines. Consequently, we found that broader and shallower cyclotron lines as commonly observed in accretion-powered pulsars are formed in such a line-forming region composed of a large number of domains. The resulting cyclotron lines also tend to have an asymmetric profile that is shallower toward lower energies due principally to the gradient in the widths of the original lines. We also find that the cyclotron absorption line can become broader, as it becomes deeper, regardless of the geometries, i.e., slabs or cylinders as a result of a superposition of individual lines.

*Subject headings:* stars: atmospheres — stars: magnetic fields — stars: neutron

### 1. INTRODUCTION

Cyclotron lines in X-ray spectra provide only direct measure of the magnetic field strength of a line-forming region in a neutron star. They also offer information on the geometry of an accretion column and the properties of a line-forming region in a neutron star. Considerable effort has been spent in attempting to reproduce the observed cyclotron features in the line-forming regions with slab or cylindrical geometries (Nagel 1981a, 1981b; Meszaros & Nagel 1985; Wang et al. 1988, 1989; Isenberg et al. 1998; Araya & Harding 1999, 2000). The profiles of observed cyclotron features in accreting X-ray pulsars, however, tend to be broader and shallower than those calculated theoretically in single-slab or cylindrical geometries.

Bulik et al. (1992) compared the pulse phase-dependent spectra of 4U 1538–52 obtained from *Ginga* to a series of static model slabs of curved accreting polar caps. As a result, they found that even maximal pole-on Doppler broadening is unable to fit the line profiles with a single field strength or a dipole distribution at constant radius. Bulik et al. (1995) also fitted to the spectra with a two-component magnetic field model. The field strength difference between the high and the low components is a factor of 5 or larger, i.e., much larger than that expected from a simple dipole field variation along the stellar surface inside the polar cap dimensions.

Burnard et al. (1991) calculated the polarimetric spectra expected from a mound of accreting plasma settling subsonically onto the polar cap regions of X-ray pulsars, representing the overall accretion mound by a series of one-dimensional slabs of non-uniform material. They found that the average of intensity over the surface of the mound greatly reduces the variation in intensity and polarization found in single-slab models. However, the effect of cyclotron resonant scattering is not included.

Becker (1998) and Basko & Sunyaev (1976) have considered the dynamics of radiation-dominated pulsar accretion flows, including the effect of the shock and the escape of radiation energy

from the column. They derived a simple analytical solution for a velocity profile in a radiation-dominated pulsar accretion column for an infinitely strong shock. Becker (1998) also shows the height of the shock above the surface of the neutron star. Furthermore, the analytical solution for the velocity provides a very accurate description of the flow structure between the location of the shock and the surface of the neutron star.

The variation on the centroid energies of the cyclotron lines has been observed with pulse phase or luminosity (Mihara et al. 2004; Nakajima et al. 2006) for the binary X-ray pulsar X0115+63. Mihara et al. (2004) reported that the first and second harmonic cyclotron features ( $\sim 11.3$  and  $\sim 22.1$  keV) were observed in the spectrum of the 1990 February flare of X0115+63 with the *Ginga* Large Area Counter, while only one feature ( $\sim 15.6$  keV) was observed in the 1991 April flare. They interpreted the change as a variation of height of an accretion column threaded by a dipole field with the luminosity. Nakajima et al. (2006) found that the fundamental resonance energy increased from  $\sim 11$  to  $\sim 16$  keV as the source luminosity  $L_X$  decreased. They concluded that these results could be explained by the changes in the accretion column height with the luminosity as theoretically predicted by Burnard et al. (1991).

Nakajima et al. (2006) reported that the increase in the ratio of the width of the absorption feature to the resonance energy can result from the luminosity-correlated elongation in the accretion column, which would cause a larger range of magnetic field strengths to take part in the line formation. Mowlavi et al. (2006) also pointed out the possibility that from the change of the observed line width in the high-mass X-ray transient V0332+53 with *INTEGRAL*, the extent of the line-forming regions can change with the accretion rate and the line-forming regions can extend over some fraction of the accretion column. They suggest that the line-forming region would cover some range of magnetic strengths. Consequently, broad-line shape would result from the superposition of a series of narrow lines that originate from a different height with a different magnetic field strength in the accretion

column. This result motivated us to perform a theoretically detailed study of the effect of the extended line-forming regions with the growth of an accretion column on the properties of the cyclotron lines. In addition, Tsygankov et al. (2006) reported that there is the possibility that the ratio of the energies of the second and main harmonics slightly decreases with decreasing luminosity in the observations of the transient X-ray pulsar V0332+53 with *INTEGRAL* and *RXTE*, although they cannot make a final conclusion about the rate of change of the second harmonic energy with the source luminosity due to lower statistics of the data at the high energies.

Nishimura (2005) calculated the cyclotron lines assuming the strong variation on the field strength with a distance from the emission region. His model can account for the deviation from classical harmonic ratios in cyclotron absorption lines. However, he cannot reproduce enough broad and shallow line profiles required for the case of accretion-powered X-ray pulsars, at least except for a 1–0 slab geometry with the field strength increasing with the distance from source plane, although he took into account the effect of some sorts of the variations on the field strength different from a dipole field.

In this work, a line-forming region is represented by a series of a large number of uniform domains at a temperature, density, and magnetic field strength that vary with the height of each of the individual domains for a cylinder geometry but a uniform density over a whole line-forming region for a slab geometry. We therefore study the properties of the spectra formed by superposing the cyclotron lines emerging from some domains. In the present paper we study the formation of the cyclotron lines by considering a configuration of an extended line-forming region composed of some slabs or cylinders. The paper is organized as follows. In § 2 we discuss configurations of some line-forming regions we use in our calculations. The radiative transfer calculations are briefly explained in § 3. In § 4 we present the spectra calculated by our model and discuss the influence of a superposition of some lines on the properties of resulting cyclotron lines. We compare our results to observations for accretion-powered pulsars in § 5. In § 6 we summarize our results.

## 2. LINE-FORMING REGION MODEL

Most efforts to calculate the theoretical cyclotron line have been performed in a uniform line-forming region at a temperature, magnetic field, and density. However, the nonuniformity in the line-forming region can affect the property of the cyclotron line, as Nishimura (2003, 2005) found that the variations on the magnetic field strength affect the profile of the cyclotron line and its harmonic ratio. We must couple a radiative transfer with a radiation hydrodynamic of the accretion flow to calculate theoretically a cyclotron line in a completely self-consistent way. In the present paper, in order to avoid such an intractable multi-dimensional time-dependent task, we employ an analytical solution for the velocity profile given by Becker (1998) and Basko & Sunyaev (1976), which considered the steady, plane-parallel flow of a radiation-dominated ideal fluid along a magnetically confined accretion column. We consider a nonuniform line-forming region composed of a large number of uniform line-forming domains in a temperature, magnetic field, and density. The density at a given height  $h$  in a line-forming region is determined by the analytical velocity profile. Our primary goal is to study the effect of a superposition of some cyclotron lines emerging from different parts of an accretion column or mound on the properties of the resulting lines. Once the temperature, the density, and the magnetic field strength in each domain are determined, we can calculate cyclo-

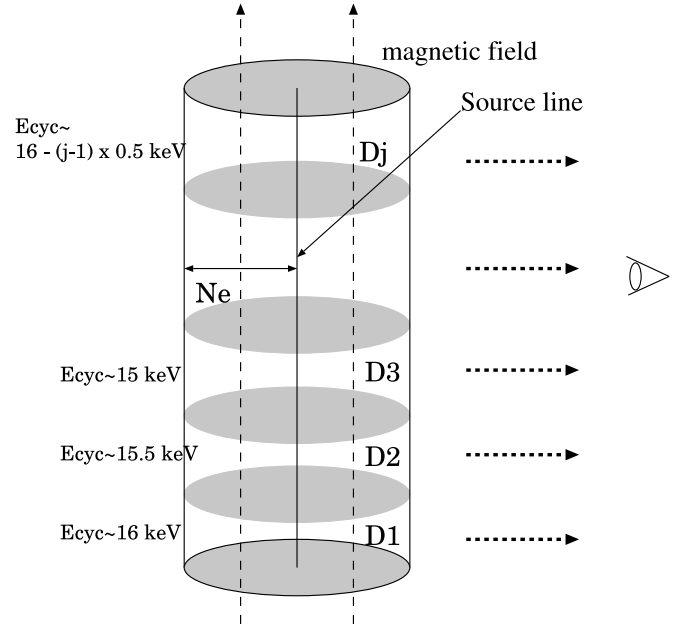


FIG. 1.—Schematic of the line-forming region composed of a large number of individual cylindrical geometries,  $D_j$  ( $j = 1, 2, 3, \dots, j_{\max}, j_{\max} \equiv 17$ ) at a temperature, density, and magnetic field strength that vary with the height of each individual domain. The magnetic field strength at the neutron star surface is  $B_s = 1.6 \times 10^{12}$  G. The resulting spectrum is computed by the sum of some spectra emerging from some individual domains. Each cylindrical geometry is threaded by the magnetic field parallel to the cylinder axis. The photon source is located along the cylinder axis.  $N_e$  is the column depth between the photon source and the surface.

tron lines emerging from each of the domains with a magnetized Feautrier radiative transfer code.

Basko & Sunyaev (1976) found a critical luminosity ( $L^* \sim 10^{37}$  ergs s $^{-1}$ ) dividing the accretion column into two cases. For high accretion rates  $L > L^*$ , the influence of the radiation is significant. In this case, the infalling gas loses practically all its kinetic energy, which transforms into radiation, in the radiative shock. In addition, only a small fraction of the accretion energy can be radiated through the side walls in the shock zone, since at  $L \gg L^*$  the transverse optical depth of the accretion channel is much greater than 1. Thus, the shock rises above the surface of the neutron star as the accretion rate  $L > L^*$  grows and an extended sinking zone appears below the shock. The accretion energy principally goes into the extended sinking zone and is gradually emitted by the side walls of the accreting column. This motivates us to consider an elevated cylindrical geometry threaded by a magnetic field varying with a height for a line-forming region. Thus, we first consider a simple static cylindrical geometry illustrated schematically in Figure 1 for a line-forming region with the surface magnetic field  $B_s = 1.6 \times 10^{12}$  G. We consider a cylinder geometry threaded by a dipole magnetic field oriented parallel to the cylinder axis over the whole region that corresponds to the postshock settling zone. We assumed that the whole line-forming region is composed of a large number of individual uniform domains,  $D_j$  ( $j = 1, 2, 3, \dots, j_{\max}$ ), at a temperature, density, and magnetic field strength that vary with the height of each domain. Each domain is assumed to be located in the height of the accretion column in steps of  $\sim 0.5$  keV in the cyclotron resonance energy  $\hbar\omega_{\text{cyc}}$  derived from the dipole field,  $B = B_s[R_*/(R_* + h)]^3$ , where  $h$  is the height of each individual domain  $D_j$ ,  $R_*$  is the stellar radius, and  $B_s$  is the magnetic field strength at the neutron star surface. The temperature at each individual domain is also determined by  $kT \sim \hbar\omega_{\text{cyc}}/4$ , which is derived by Lamb et al. (1990)

from a balance of the cooling and heating that arise from cyclotron resonant and nonresonant scattering. The profile of the flow velocity  $v(h)$  in the accretion column is given by

$$v(h) = -v_{\text{ff}} \left[ 1 - \left( \frac{7}{3} \right)^{-h/h_{\text{shock}}} \right], \quad (1)$$

where  $v_{\text{ff}} \equiv (2GM_*/R_*)^{1/2}$  is the free-fall velocity from infinity onto the surface of a neutron star and  $M_*$  is the stellar mass. The mass accretion rate  $\dot{M}$  is also related to the site of the shock via

$$\dot{M} = \frac{3m_p c \pi}{\sigma_T} \frac{h_{\text{shock}}}{\ln(7/3)}. \quad (2)$$

In our calculations for the cylinder geometry, we therefore take the density profile in each domain at the height  $h$  from the surface of a neutron star to be

$$\rho(h) = \frac{\dot{M}}{\pi r_0^2 v_{\text{ff}} \left[ 1 - (7/3)^{-h/h_{\text{shock}}} \right]}, \quad (3)$$

which is derived by substituting equation (1) into the continuity equation,  $\dot{M} = \pi r_0^2 \rho(h) |v(h)| = \text{const}$ . In the cylinder geometry, the polar cap radius is  $r_0 = 10^5$  cm and the stellar mass  $M_* = 1.4 M_\odot$  as the general case for neutron stars. The shock height  $h_{\text{shock}}$  is a free parameter.

For low luminosity  $L < L^*$ , the influence of the radiation on the falling matter is negligible. In this case, the gas falls freely right down to the surface of the neutron star, where it loses its kinetic energy through collisions with electrons and protons of the neutron star atmosphere. This motivates us to consider a slab geometry with a different magnetic field, e.g., higher order multipoles, for a line-forming region. Thus, we consider a line-forming region represented by a group of the plane-parallel 1–0 slab model, which is a slab illuminated from below as defined in Isenberg et al. (1998). Here a magnetic field is assumed to be parallel to the slab normal. As in the cylinder geometry, we assumed that the whole line-forming region is composed of a large number of individual uniform domains,  $D_j$  ( $j = 1, 2, 3, \dots, j_{\text{max}}, j_{\text{max}} \equiv 17$ ), at a temperature, magnetic field strength, and density. The whole line-forming region, which is composed of a number of uniform domains at the temperature, magnetic field, and density, is assumed to be a nonuniform temperature and magnetic field strength but a uniform electron number density. This model may correspond to a line-forming region where a large number of the slabs with a disorganized field distribution are set horizontally, i.e., the case of flow stopping through Coulomb collisions with the ambient electrons in a slab, e.g., with random or chaotic small-scale variations of the surface field, higher order multipoles at the same height (e.g., Bulik et al. 1992, 1995), or maybe a larger polar cap area ( $r_0 > 10^5$  cm) with a dipole field. This line-forming region may also be considered as the accretion mound with a dipole field like in Burnard et al. (1991), although for each domain we neglect the inclination and azimuth of each individual single slab with respect to the surface of the star. Each of the slab domains is assumed to be threaded by the magnetic field determined by steps of  $\sim 0.5$  keV in the cyclotron resonance energy  $\hbar\omega_{\text{cyc}}$ , and its temperature is also determined by  $kT \sim \hbar\omega_{\text{cyc}}/4$ . A cyclotron line feature for the angle  $\mu = 0.57$ , which is principally used in our works, in the single 1–1 slab geometry should be similar to one in the single cylinder geometry as shown in Araya & Harding (1999). The resulting spectra in the 1–1 slab

geometry are therefore expected to resemble those in the cylinder geometry so that we only consider the 1–0 slab geometry for a slab.

In our works, each domain in the line-forming region is assumed to be independent. This assumption may not hold especially for the photon propagating at a small angle to the magnetic field line ( $\mu \sim 1$ ), since the photon at the small angle can pass through some other domains before escaping from the line-forming region. However, our calculations are principally performed with the angle  $\mu = 0.57$ . The line formation is also expected to have mainly occurred near the outer edge of the accretion column. Moreover, the photons traveling some domains principally tend to render each original line feature somewhat broader and shallower (Nishimura 2005). Thus, this assumption would not influence significantly the resulting spectra and our conclusions. Although this model may also be insufficient to express an actual accretion column or mound on the surface of a neutron star, we would be able to examine the properties of the superposition of some line features formed in some individual domains with the different field strengths in the cylinder or slab geometry and offer guides to understanding the line features formed in an actual accretion column or mound.

### 3. RADIATIVE TRANSFER

We use a difference method to solve radiative transfer, the so-called Feautrier method (Mihalas 1978, pp. 151–158). The cyclotron line is modeled by superposing some of the lines calculated in the individual domains with a Feautrier code with 100 frequencies, one angle or 60 frequencies, four angles: the total photon density is computed as the sum,  $U_{\text{total}} = \sum_{i=1}^J U_{D_i}$ , of some photon densities,  $U_{D_i}$ , emerging from the individual domains. An initial power-law photon density,  $U_i(E) \propto \omega^{-1}$ , is injected into each individual domain. Wang et al. (1988) found that polarized transfer with the vacuum modes is very similar to unpolarized transfer. This is because the perpendicular and parallel modes are strongly coupled and each contribute about equally to the emergent frequency spectra and angular distributions. The polarization-averaged cross sections therefore are suitable for first harmonic scattering in optically thick media when the vacuum contribution to the dielectric tensor dominates the plasma contribution, i.e.,

$$\frac{w}{\delta} = 3 \times 10^{-6} \left( \frac{N_e}{10^{23} \text{ cm}^{-3}} \right) \left( \frac{B}{B_{\text{crit}}} \right)^{-4} \ll 1, \quad (4)$$

where  $w \equiv (\omega_p/\omega_{\text{cyc}})^2$  is the plasma frequency parameter,  $\omega_p$  is the plasma frequency,  $\delta$  is the magnetic vacuum polarization parameter (Adler 1971), and the critical magnetic field  $B_{\text{crit}} = 4.414 \times 10^{13}$  G. When the electron number density  $N_{e,21} \sim 0.1$ –100, as we discuss in this paper, the line-forming region may be optically thick to line photons but will be optically thin to continuum photons, since the optical depths in the cyclotron fundamental and the continuum are given by  $\tau_1 = 100 N_{e,21} B_{12}^{-1} (T/\text{keV})^{-1/2}$  and  $\tau_c \sim \sigma_T N_{e,21} = 6.7 \times 10^{-4} N_{e,21}$ , respectively. Here  $N_e \equiv N_{e,21} \times 10^{21}$  electrons  $\text{cm}^{-2}$  and  $B = B_{12} \times 10^{12}$  G (Lamb et al. 1990). In the present work, this condition generally holds so that the continuum scattering would not have significant influence on our conclusions. Thus, a polarization-averaged cross section is employed and the continuum scattering is approximated by Thomson scattering in this work. We also employ the cold plasma approximation in which  $kT \ll \hbar\omega_{\text{cyc}}$ : the Landau level spacing is much larger than typical electron energy. Thus, the Landau level is not collisionally populated. Moreover, the levels are not radiatively populated since the photon densities are assumed to

be sufficiently low. Consequently, the initial and final electron state is the Landau ground state ( $n = 0$ ) in each resonant scattering. Most of the electrons that are excited to states  $n \geq 2$  de-excite to the ground state through intermediate excited levels for the magnetic field strengths considered here, so that spawned photons refill the fundamental line. We include excitation up to three harmonics in Landau levels and photon spawning followed by decay of these levels in cyclotron resonant scattering.

We use the complete redistribution  $\phi(x, \mu)\phi(x', \mu')$  for two-stream approximation and the redistribution  $R(x', \mu'; x, \mu)$  derived by Wasserman & Salpeter (1980) for multistream calculations with the four angles  $\mu = 0.18, 0.53, 0.79$ , and  $0.96$  chosen by a Gauss quadrature formula. Here  $\phi(x, \mu)$  is the nonrelativistic polarization-averaged scattering profile and  $x$  is given by  $x = (\omega - \omega_{\text{cyc}})/\omega_{\text{cyc}}(2kT/m_e c^2)^{1/2}$ . The angles are represented by  $\mu = \cos \theta$ , where  $\theta$  is the photon propagation angle relative to the magnetic field. We take the distribution of the electron momenta along the field,  $f(p)$ , to be a one-dimensional nonrelativistic Maxwellian,  $f(p) dp = N_e [\exp(-u^2)/\sqrt{\pi}] du$ , where  $u = p/(2m_e T)^{1/2}$ . The effect of the relativistic scattering profile is not included because its effect is relatively small for the angle  $\mu = 0.57$  and the magnetic field strength  $B \ll B_{\text{crit}}$  used mainly in our calculations, although relativistic kinematics significantly affects the shape of the scattering profile for small  $\mu$  even in the limit  $E, kT \ll m_e c^2$  (Lamb et al. 1989). A nonrelativistic resonant scattering cross section at the  $n$ th harmonic for  $n \geq 1$ , which is valid for  $B \ll B_{\text{crit}}$ , is given by (Harding & Daugherty 1991)

$$\sigma_{\text{res}}^n = \frac{\alpha \pi^2 \hbar^2}{m_e} \left( \frac{B}{B_{\text{crit}}} \right)^{n-1} \frac{(n^2/2)^{n-1}}{(n-1)!} \times (1 + \cos^2 \theta) \sin^{2(n-1)} \theta \frac{\Gamma_n/2\pi}{(\omega - \omega_n)^2 + \Gamma_n^2/4}. \quad (5)$$

Here the width  $\Gamma_n$  is the decay rate from the Landau excited level  $n$ . The free-free absorption is also negligible, since the density is assumed to be sufficiently low compared to  $N_{e,21}^{\text{th}} \sim 4.2 \times 10^4 (kT_e/4 \text{ keV})^{1/3} (\hbar\omega/16 \text{ keV})^{7/6}$ , where photons with energy  $\hbar\omega$  will be thermalized prior to escape (Isenberg et al. 1998; Nelson et al. 1993).

In order to discuss strictly the harmonic ratios in cyclotron line energies, we use the exact relativistic cyclotron resonant energy given by

$$\hbar\omega_n = \frac{mc^2 \left\{ [1 + 2n(B/B_{\text{crit}}) \sin^2 \theta]^{1/2} - 1 \right\}}{\sin^2 \theta}. \quad (6)$$

We also employed the relativistic transition rate to the ground state from the  $n$ th harmonic given by equation (27) in Latal (1986). In the present paper we refer to the wing at lower energies than the line center as the red wing and the wing at higher energies than the line center as the blue wing.

## 4. RESULTS

### 4.1. Cylinder Geometry

The whole column is represented by a set of a large number of domains with cylindrical geometry as shown in Figure 1. The resulting cyclotron lines are modeled by computing the sum of some lines emerging from some individual domains.

#### 4.1.1. Optically Thin Case

Computations were carried out with a two-stream approximation. The magnetic field strengths at the neutron star surface  $B_s$

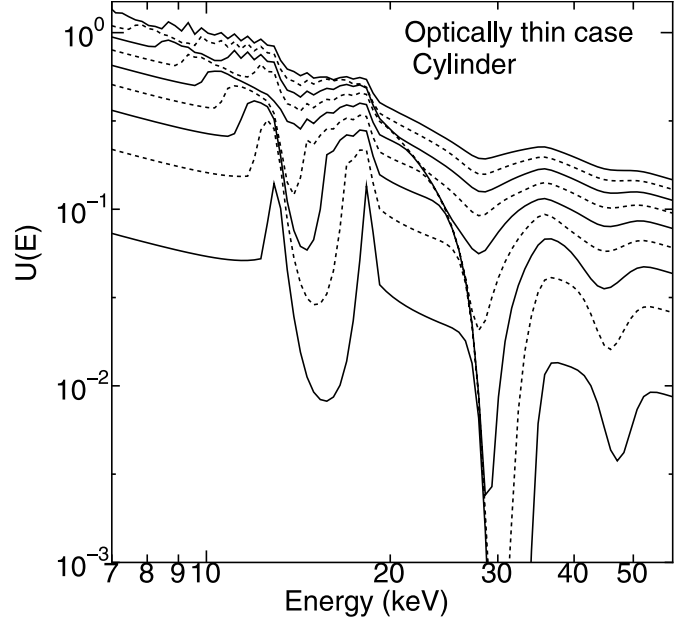


FIG. 2.—Spectra generated by the superposition of some cyclotron lines emerging from each of the individual domains in cylindrical geometry for the following eight resonance energy bands: 8–16, 9–16, 10–16, 11–16, 12–16, 13–16, 14–16, 15–16 keV (top to bottom). The bottom spectrum is in a single domain with the cyclotron energy  $E_{\text{cyc}} = 16$  keV. The electron number density,  $N_{e,21}$ , varies from 1 to 5.7 over the top through the bottom of the whole line-forming region.

are taken to be  $1.6 \times 10^{12}$  G, which was one of two values considered in Nakajima et al. (2006) for X0115+63, and the maximum number of the domain  $j_{\text{max}}$  is set to be 17. Figure 2 shows the spectra computed by the superposition of some individual narrow cyclotron lines in the line-forming region with the cyclotron resonance energies in the range  $\Delta E_{\text{cyc}} = 8\text{--}16$  keV spaced every  $dE \sim 0.5$  keV and the column depths in the range from  $N_{e,21} = 1$  to 5.7. The shock height  $h_{\text{shock}}$  is taken to be  $3.3 \times 10^5$  cm, which corresponds to the height with the cyclotron energy  $\sim 8$  keV. Here we study how the structure of the cyclotron lines can change with the number of the superposition of the lines, fixing the shock height, i.e., the luminosity.

The lowest solid line indicates the cyclotron line emerging from the single uniform cylindrical domain at a temperature, density, and magnetic field, the strength of which corresponds to resonance energy  $\sim 16$  keV. The individual original cyclotron line features are even deeper and narrower as theoretically calculated by a large number of previous works (e.g., Isenberg et al. 1998; Araya & Harding 1999, 2000) in a single cylinder geometry with  $N_e \sim 10^{21}$  electrons  $\text{cm}^{-2}$ . However, the superposition of the individual deeper and narrower lines leads to a shallower and broader cyclotron line. Each original fundamental cyclotron line has the structure of one absorption-like feature with two emission-like features at both wings, while the structures of the second and third harmonic lines have only one simple absorption feature as shown in Figure 3. As the number of the superposition of the cyclotron lines increases, the shape of the fundamental line becomes more and more complex as can be seen in Figure 2. This is because each individual fundamental line is composed of one absorption-like feature and two emission-like features at both sides of the line center. The fundamental line profile also becomes broader and shallower because the superposition of a large number of original narrow cyclotron lines, which are composed of one absorption-like feature and two emission-like wings with different peak energies, suppresses both the absorption

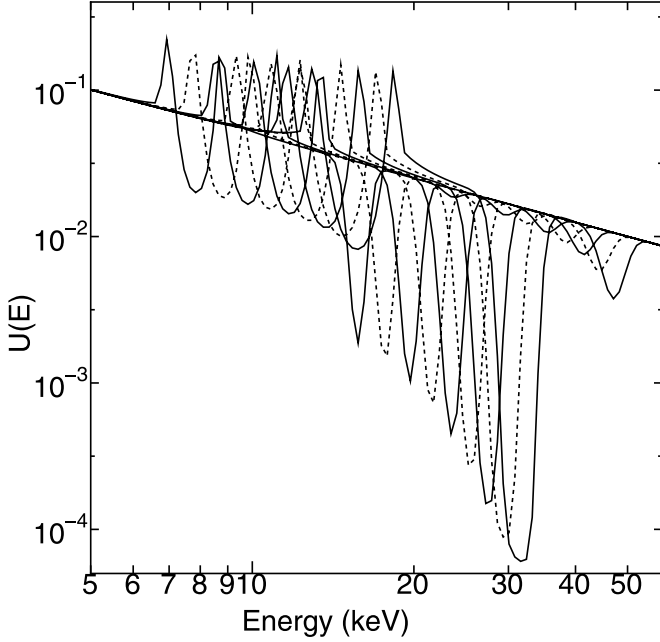


FIG. 3.—Original individual cyclotron line emerging from each of the line-forming domains. The spectra show the original lines with the cyclotron resonance energies from 8 to 16 keV in steps of 1 keV. The fundamental lines have the structure of an absorption-like feature with two emission-like features at both wings, while the structures of the second and third harmonic lines have only a simple absorption feature.

and emission features of each original line. Consequently, the absorption-like feature is broader and shallower and the emission-like features at both wings are less prominent. In the region of the cyclotron energy ranges from  $\Delta E_{\text{cyc}} = 12\text{--}16$  to  $10\text{--}16$  keV, the fundamental line appears to consist of double absorption features due to the superposition of a large number of cyclotron lines with different peak energies. The resulting cyclotron line profiles are broader, shallower, and more complex at the fundamental than those in a single cylindrical model. It may become more and more difficult to detect this feature as a line, as the number of the superposition of the cyclotron lines increases.

On the other hand, the second and third harmonic lines also become broader and shallower, keeping one clear absorption structure in contrast to the complex fundamental feature, since the shape of each individual second and third harmonic line has only one absorption-like feature without emission-like wings. However, the second and third harmonic lines also have an asymmetric shape that is shallower toward lower energies. This is due principally to the two reasons below: First, each original line becomes narrower as the cyclotron resonance energy decreases, since the thermal Doppler width diminishes due to the decrease of the plasma temperature via  $kT \sim \hbar \omega_{\text{cyc}}/4$ . As a result, an overlapping area of each line diminishes toward lower energy. Second, the density profile in the accretion column decreases with height. Thus, the absorption feature becomes shallower toward lower energy. Thus, the peaks of all harmonic absorption lines tend to indicate relatively higher ones of the cyclotron resonant energies incorporated in the whole line-forming region. However, the deviation from a Gaussian profile at the second and third harmonics will be expected to be smaller than at the fundamental, since the original lines of the second and third harmonics do not have emission wings so that the resulting line profile cannot be as complex as the fundamental. Thus, the line profile at higher harmonics may be able to be approximated by a Gaussian profile.

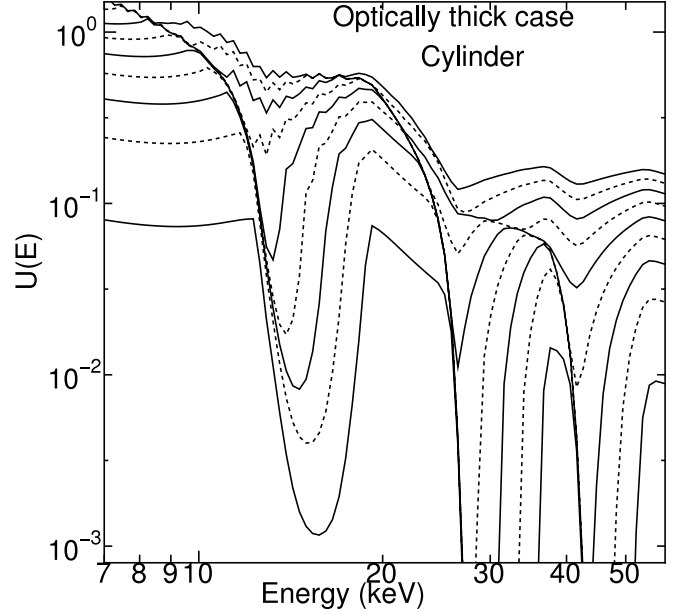


FIG. 4.—Same as Fig. 2, but the electron number density,  $N_{e,21}$ , varies from 20 to 114 over the top through the bottom of the whole line-forming region.

#### 4.1.2. Optically Thick Case

Figure 4 is the same as Figure 2, but for the range of column depths from  $N_{e,21} = 20$  to 114. Each original fundamental absorption-like feature is even deeper than that in the optically thin case ( $N_{e,21} = 1\text{--}5.7$ ). The resulting fundamental lines therefore tend to indicate the absorption-like features with relatively small emission-like wings in this optical depth, as already noted by Isenberg et al. (1998). The emission-like wings become less prominent with the increase of the optical depth so that the cyclotron profile is similar to that in the 1–0 slab geometry. This line feature is likely interpreted as rather absorption-like than emission-like as observed commonly in the accretion-powered X-ray pulsar spectra. The original line depth becomes narrower and shallower with the height of the line-forming domain due to the gradients in the temperature, density, and cyclotron energy with the height of the domain as shown in Figure 3, so that the resulting fundamental line becomes shallower toward lower energies. Consequently, the peak of the resulting cyclotron absorption line indicates somewhat higher ones of some cyclotron frequencies incorporated in the overall line-forming region. The profile of the cyclotron absorption line thus becomes obviously asymmetric.

#### 4.1.3. Higher Cyclotron Energy

Figure 5 shows the spectra formed by the superposition of a large number of the cyclotron lines in the range of the cyclotron energy  $\Delta E_{\text{cyc}} = 17\text{--}30$  keV in the cylinder geometry with the range of column depths from  $N_{e,21} = 45$  to 120. The surface magnetic field  $B_s$  is taken to be  $2.6 \times 10^{12}$  G, and the maximum number of the domain  $j_{\text{max}}$  is set to be 27. The line energy shifts from 29 to 22 keV as the number of the superposition of the lines increases and then stays around  $\sim 22$  keV. The Doppler width is given by  $\omega_{\text{cyc}}(2kT/mc^2)^{1/2}|\cos \theta|$  so that the line width becomes broader with the cyclotron energy and the plasma temperature. A clear absorption line continues to appear until the range of the cyclotron energy  $\Delta E_{\text{cyc}} = 19\text{--}30$  keV in spite of the superposition of a large number of the lines. This is because there is an overlapping area over a large number of the absorption lines,  $\Delta E_{\text{cyc}} = 19\text{--}30$  keV, due to broader line structures. Thus, a clear absorption line continues to survive even when the superposition

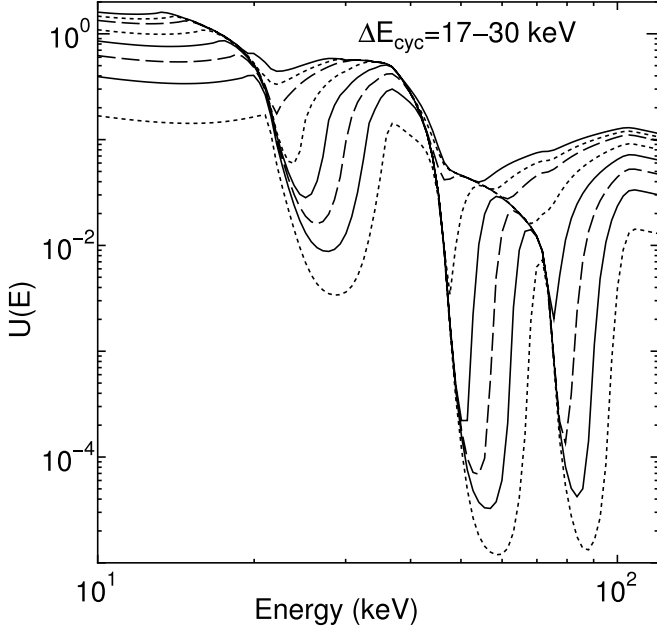


FIG. 5.—Same as Fig. 4, but for the spectrum superposed over the ranges of the cyclotron energy 17–30, 18–30, 19–30, 21–30, 23–30, 25–30, 27–30, and 29–30 keV (top to bottom). The electron number density,  $N_{e,21}$ , varies from 45 to 120 over the top through the bottom of the whole line-forming region.

of the lines occurs, where the line energy continues to shift to lower energies until  $\sim 22$  keV. The behavior of the variation on the line energy with the number of the superposition of the lines is therefore dependent on the line width, i.e., the line energy and the plasma temperature and so on.

#### 4.1.4. Luminosity-dependent Spectra

Figures 6 and 7 show the variation on the spectra with the luminosity for the ranges of the column depths from  $N_{e,21} = 1$  to 5.7 and the ranges from  $N_{e,21} = 4.7$  to 114, respectively. The luminosity is related to  $h_{\text{shock}}$  via the following equation (Becker 1998):

$$L_X = \frac{3GM_* m_p c \pi}{\sigma_T \ln(7/3)} \left( \frac{h_{\text{shock}}}{R_*} \right). \quad (7)$$

The density at a given height,  $h$ , is determined by

$$\rho(h) = \frac{R_* L_X}{GM_* \pi r_0^2 \left[ v_{\text{ff}} \left( 1 - (7/3)^{-h/h_{\text{shock}}} \right) \right]}, \quad (8)$$

derived by substituting  $\dot{M} = (R_*/GM_*)L_X$  into equation (3). Here we assumed the upstream (incident) Mach number in far upstream region  $\mathcal{M}_\infty \rightarrow \infty$ , which is an excellent approximation in pulsar accretion flows (Basko & Sunyaev 1975, 1976).

For the ranges of the column depths from  $N_{e,21} = 1$  to 5.7, the properties of the spectra are generally similar to those calculated in the case of constant  $h_{\text{shock}}$  in the luminosity from  $L_X = 3.1 \times 10^{37}$  to  $5.1 \times 10^{37}$  ergs s $^{-1}$  (from 8–16 to 11–16 keV), while we can see some differences in the line depth from  $L_X = 9.3 \times 10^{36}$  to  $2.6 \times 10^{37}$  ergs s $^{-1}$  (from  $\Delta E_{\text{cyc}} = 16$  to 12–16 keV), where the line depth at the fundamental is shallower than those in Figure 2. The density is proportional to the luminosity as is evident from equation (8), so that the depth of the line does not decrease monotonically with increasing luminosity while the superposition of the lines reduces the line depth. In the luminosity  $L_X = 1.3 \times$

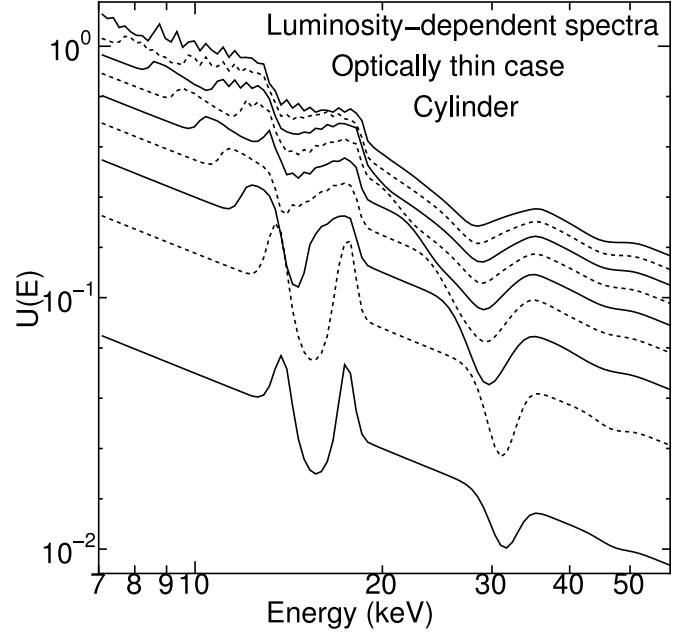


FIG. 6.—Spectrum of cylindrical columns with increasing luminosity  $9.3 \times 10^{36}$ ,  $1.3 \times 10^{37}$ ,  $1.7 \times 10^{37}$ ,  $2.1 \times 10^{37}$ ,  $2.6 \times 10^{37}$ ,  $3.1 \times 10^{37}$ ,  $3.7 \times 10^{37}$ ,  $4.3 \times 10^{37}$ , and  $5.1 \times 10^{37}$  ergs s $^{-1}$  (bottom to top). The electron number density,  $N_{e,21}$ , varies from 0.23 to 5.7 over the top through the bottom of the whole line-forming region.

$10^{37}$  ergs s $^{-1}$  ( $\Delta E_{\text{cyc}} = 15$ –16 keV), the three cyclotron lines become first deeper than those in a single domain with the cyclotron resonance energy 16 keV ( $L_X = 9.3 \times 10^{36}$  ergs s $^{-1}$ ), in spite of the superposition of the three spectra. Then, the line depth gradually decreases with increasing luminosity from a balance of the deepening effect due to the increase of the density with the luminosity and the shallowing effect due to the superposition of the lines until  $L_X = 2.1 \times 10^{37}$  ergs s $^{-1}$  ( $\Delta E_{\text{cyc}} = 13$ –16 keV), being almost constant from  $L_X = 2.6 \times 10^{37}$  to  $5.1 \times 10^{37}$  ergs s $^{-1}$  (from  $\Delta E_{\text{cyc}} = 12$ –16 to 8–16 keV). On the other hand, the line width first decreases and then tends to increase with the luminosity.

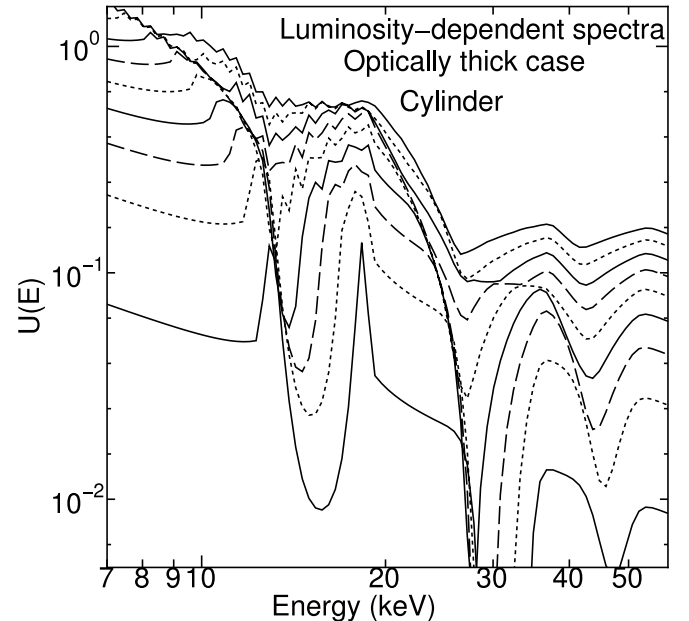


FIG. 7.—Same as Fig. 6, but the electron number density,  $N_{e,21}$ , varies from 4.7 to 114 over the top through the bottom of the whole line-forming region.

Moreover, the double absorption features begin to appear even in the luminosity  $L_X = 2.6 \times 10^{37}$  ergs  $s^{-1}$  ( $\Delta E_{cyc} = 12\text{--}16$  keV) due to the increase of the density.

For the ranges from  $N_{e,21} = 4.7$  to 114, the line depth decreases with increasing luminosity, once it increases until  $L_X = 1.7 \times 10^{37}$  ergs  $s^{-1}$  ( $\Delta E_{cyc} = 14\text{--}16$  keV) due to the increase of the density in the line-forming region with the luminosity. However, the line depth seems to be almost constant from  $L_X = 1.3 \times 10^{37}$  to  $2.1 \times 10^{37}$  ergs  $s^{-1}$  (from  $\Delta E_{cyc} = 13\text{--}16$  to  $15\text{--}16$  keV). The line width generally indicates a tendency to increase with the luminosity, although it tends to decrease from the luminosity  $L_X = 9.3 \times 10^{36}$  to  $2.1 \times 10^{37}$  ergs  $s^{-1}$  (from  $\Delta E_{cyc} = 16$  to  $13\text{--}16$  keV) due to the presence of an overlapping area in a number of the absorption lines. The resulting absorption lines also gradually get to be similar to those in the 1–0 slab geometry due to the reduction of the emission wings with the increase of the optical depth. Thus, the double absorption feature, as can be seen in the ranges of the column depth  $N_{e,21} = 1\text{--}5.7$ , comes to be less prominent. In the luminosity  $5.1 \times 10^{37}$  ergs  $s^{-1}$  ( $\Delta E_{cyc} = 8\text{--}16$  keV), the shallower fundamental and deeper second harmonic lines, as sometimes observed in accreting X-ray pulsars such as X0115+63 (Santangelo et al. 1999), are formed.

The line depth does not change with the luminosity as much as that calculated in the constant luminosity. However, it first becomes deeper with increasing luminosity and then gradually shallower, since the effect of the reduced line depth due to the superposition of the lines dominates over the effect of the increase of the density with the luminosity.

#### 4.2. Slab Geometry

We consider a line-forming region represented by a group of the plane-parallel 1–0 geometry with a nonuniform temperature and magnetic field strength but a uniform density over the whole line-forming region. In this work, the electron number density is taken to be  $N_{e,21} = 5$  for the slab geometry. Each line-forming domain is assumed to be located at the same height such as Coulomb-heated atmospheres, which have a height much less than the polar cap radius, in the case of flow stopping through Coulomb collisions with the ambient electrons. This case may correspond to a slab with random or chaotic small-scale variations of the surface field or higher order multipoles (e.g., Bulik et al. 1992, 1995). This line-forming region may also be considered as the accretion mound with a dipole field like in Burnard et al. (1991).

The resulting cyclotron line is derived by the sum of the cyclotron lines calculated at a large number of domains with the 1–0 slab geometry. In this geometry, photons that emerge from the bottom of the slab are assumed to return to the stellar surface where they are thermalized. Each cyclotron fundamental line indicates only one absorption feature without emission-like wings in contrast with the cylindrical geometry. Consequently, as shown in Figure 8, the resulting cyclotron fundamental line profile becomes a simple broader and shallower absorption feature as seen in the second and third harmonic lines after its width first becomes narrower, when the number of the superposition of the cyclotron lines increases. Each original line also becomes narrower with the decrease of the cyclotron resonance energy, since the thermal Doppler width decreases via  $kT \sim \hbar\omega_{cyc}/4$ . Thus, an overlapping area of each line diminishes toward lower energy so that the fundamental line has an asymmetric profile that is shallower toward lower energies. In contrast with the cylinder, the shape of the fundamental line is similar to those of higher harmonics, since it is not influenced by the emission-like wings as seen at the original line profiles in the cylindrical geometry. As

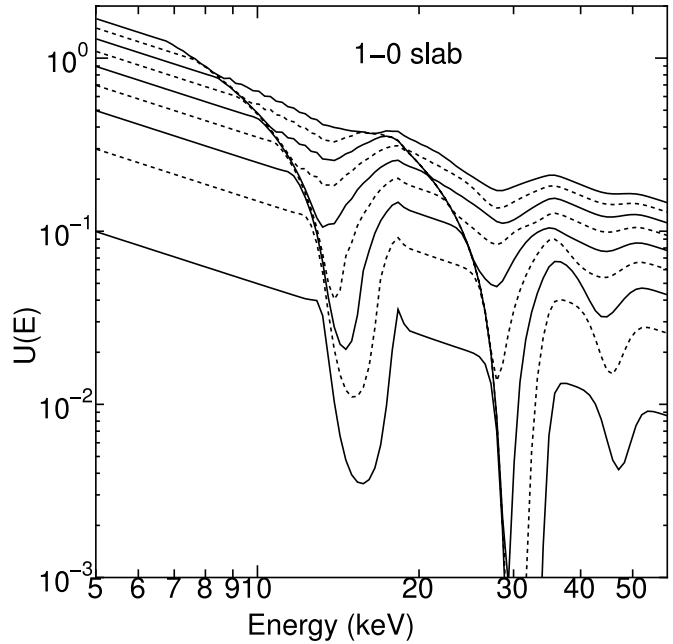


FIG. 8.—Same as Fig. 2, but in a 1–0 slab geometry with a uniform density.

a result, the structures of the cyclotron fundamental lines are simpler than those in the cylindrical geometry, although it may be difficult to fit by a Gaussian profile due to the asymmetric profile. The profiles of the second and third harmonic lines in the 1–0 slab geometry, however, are very similar to those in the cylinder geometry. This is because the shapes of the second and third harmonic lines generated through a true absorption process rather than scattering do not depend strongly on the site of the photon source.

#### 4.3. Angle-dependent Spectra

##### 4.3.1. Cylinder

Computations were carried out with a multistream Feautrier code using the four angles in the line-forming region with  $N_{e,21} = 2.2\text{--}5$ . Figure 9 shows the angle-dependent spectra modeled by accumulating the photon density emerging from a large number of cylindrical domains for the same photon's propagation angles with respect to the field. The broader lines also remain deeper at the fundamental lines like in a single cylindrical model, even when the superposition of a large number of the lines emerging from the cylindrical domains occurs. This characteristic is consistent with that observed commonly in accretion-powered X-ray pulsars (Coburn et al. 2002). However, the dependence of the higher harmonic lines on the angle is different from the fundamental, since the scattering profiles of higher harmonics have a  $\sin^{2(n-1)}\theta$  factor that leads the cross section to vanish along the field. At the smallest angle, the higher harmonic absorption lines therefore become shallowest. These characteristics remain the same as those in a single cylindrical model.

##### 4.3.2. Slab

Angle-dependent spectra are calculated in 1–0 slab domains by the same manner as in the cylinder. Figure 10 shows that at the largest angle  $\mu = 0.18$ , the narrowest and shallowest fundamental feature forms owing to the superposition of a large number of narrower lines, although each original line is deeper. Each original line is narrower due to smaller Doppler broadening so that the superposition of each original narrower line effectively

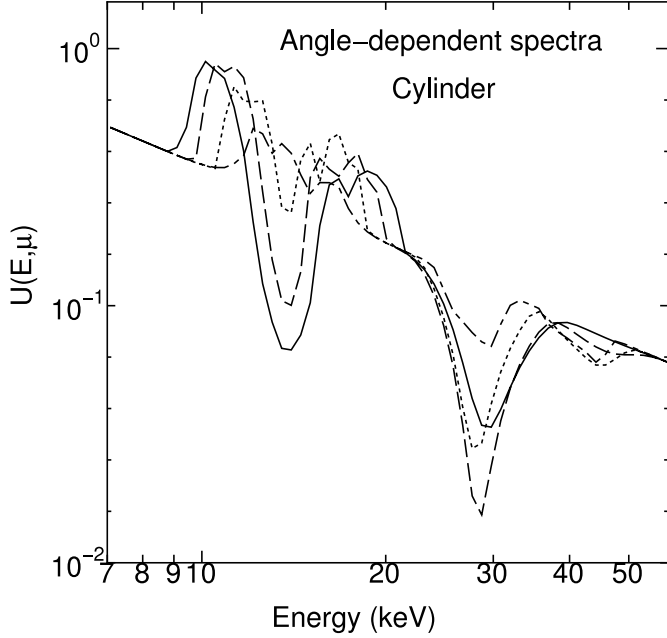


FIG. 9.—Angle-dependent spectra for four angles in cylindrical geometry. The dot-dashed, dotted, dashed, and solid lines represent the spectra for the angles  $\mu = 0.18, 0.53, 0.80$ , and  $0.96$ , respectively. The line-forming region includes the range of resonance energies 13–16 keV. The electron number density varies from  $N_{e,21} = 2.2$  to 5 over the top through the bottom of the whole line-forming region.

diminishes the depth of the resulting absorption line because an overlapping area of each original absorption feature decreases as the profile of each original line becomes narrower. Consequently, the resulting line feature is shallower at larger angle as in the case of the cylinder geometry. At the smallest angle  $\mu = 0.96$ , on the other hand, each original line is broadest and deepest owing to the largest Doppler broadening. The resulting fundamental line feature, however, becomes broader and deeper in the line-forming region composed of a large number of domains ( $\geq 7$  domains, i.e., the width of the cyclotron energy range  $\delta E_{\text{cyc}} \geq 3$  keV), since the overlapping area of each original line increases as each original line becomes broader. We can therefore see that the correlation between the depth and the width in the fundamental absorption line that is derived from the data in observations (i.e., deeper lines are also broader; Coburn et al. 2002) holds for the slab geometry as well as the cylinder. This is because the depth and width of the resulting absorption line are highly influenced by the overlapping proportion of each original line feature.

## 5. DISCUSSIONS

### 5.1. V0332+53 and X0115+63

Mowlavi et al. (2006) observed the increase of the energy and depth of the cyclotron fundamental line with time, together with the decrease of its width in the high-mass X-ray transient V0332+53, with *INTEGRAL*. Figure 5 indicates that the depth and peak energy of the line on the whole tend to decrease until a critical number  $j_{\text{crit}} \sim 23$  ( $\Delta E_{\text{cyc}} = 19$ –30 keV) of the superposition of the cyclotron lines. On the other hand, its width tends to decrease with increasing luminosity, while it increases above a critical number  $j_{\text{crit}} \sim 23$ . Our results are generally consistent with the trend of the variation in the depth of the line, as well as its energy and width in the observed cyclotron fundamental line features in V0332+53. However, the amount of the variation in

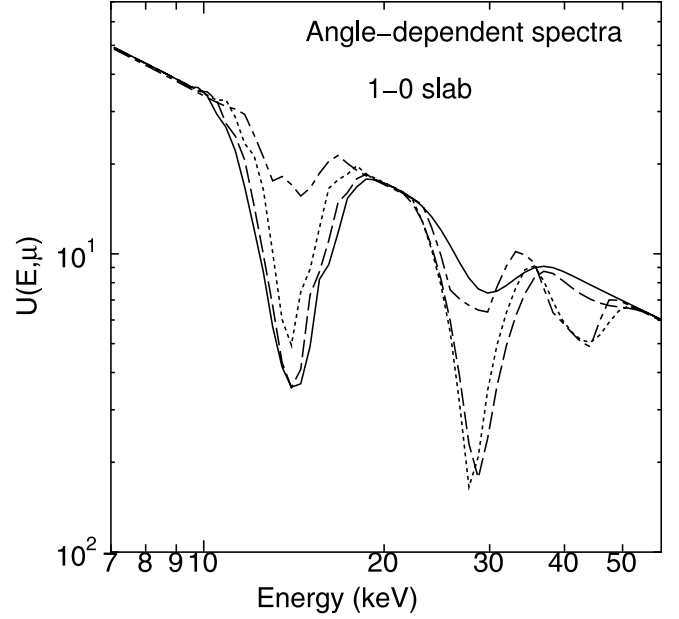


FIG. 10.—Angle-dependent spectra for four angles in the 1–0 slab geometry. The dot-dashed, dotted, dashed, and solid lines represent the spectra for the angles  $\mu = 0.18, 0.53, 0.80$ , and  $0.96$ , respectively. The line-forming region includes the range of resonance energies 13–16 keV. The electron number density of each slab is  $N_{e,21} = 5$ .

the cyclotron line energy with the luminosity in V0332+53 is too small to produce the luminosity-dependent spectra with the present model, which includes most parts of the sinking zone below the shock. This may imply the presence of the effective zone of the accretion column to the line formation.

In X0115+63, while its depth is almost constant, the same dependence of the energy of the line on the luminosity has been observed as in V0332+53 up to around the cyclotron energy  $\sim 10$  keV where the energy of the fundamental line stagnates (Nakajima et al. 2006). Figure 6 reveals that the depth and peak energy of the line on the whole tend to decrease until a critical number  $j_{\text{crit}} \sim 13$  ( $\Delta E_{\text{cyc}} = 10$ –16 keV) of the superposition of the cyclotron lines except the single domain with the cyclotron energy  $\sim 16$  keV. On the other hand, it may be difficult to determine its width above a critical number  $j_{\text{crit}} \sim 13$ , once it tends to decrease from the luminosity  $L_X = 9.3 \times 10^{36}$  to  $2.1 \times 10^{37}$  ergs s $^{-1}$  (from  $\Delta E_{\text{cyc}} = 16$  to 13–16 keV). From a critical number  $j_{\text{crit}}$ , the fundamental line width has basically a tendency to increase with the luminosity but can appear to rather decrease slightly, as shown in Figure 6. In X0115+63, the line width tends to decrease slightly with the luminosity where a resonant energy indicates around  $\sim 10$  keV. In this transition point, the number of the individual lines contributing to the line formation might be related to the critical number.

Pottschmidt et al. (2005) reported that the fundamental cyclotron resonant scattering feature (CRSF) observed in V0332+53 with *RXTE* could not be modeled well with any simple line shape. They added a shallow and narrow line component at the energy  $E_{\text{cyc}} = 25.2^{+0.2}_{-0.2}$  keV, similar to the fundamental line energy  $\sim 27.1^{+0.2}_{-0.1}$  keV, in order to produce the complex shape of the fundamental cyclotron line. Heindl et al. (1999) also reported that the fundamental line is very complex and decidedly non-Gaussian in shape in X0115+63. They required two overlapping lines to model this line, while the four higher harmonics are much more Gaussian in shape. The cyclotron fundamental line



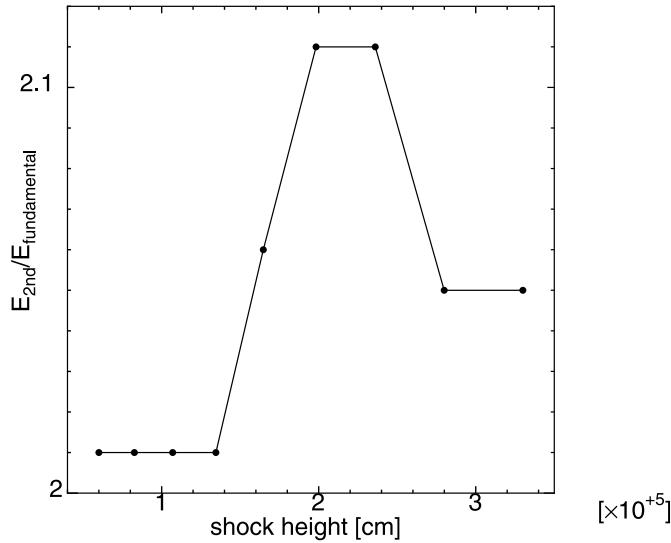


FIG. 11.—Ratios of the peak energy of the cyclotron absorption line of the second to the fundamental as a function of the shock height for cylindrical geometry using one angle,  $\mu = 0.57$ . The harmonic ratio generally tends to increase with the height of the shock, i.e., the luminosity, except for the shock heights  $2.8 \times 10^5$  and  $3.3 \times 10^5$  cm (for cyclotron energy ranges 8–16 and 9–16 keV, respectively).

profile formed by the sum of the photon densities emerging from the individual domains with the cylindrical geometry can be so complex that some fundamental lines can appear to consist of double absorption features for the width of the cyclotron energy range  $\delta E_{cyc} \sim 4\text{--}6$  keV as seen in Figure 6. As a result, the observed spectra may be able to be fitted by a two-component model. On the other hand, the second and third harmonic lines are much simpler than the fundamental even in the case of cylindrical geometry, since their original lines are composed of one absorption feature without emission-like wings. Such a complicated fundamental line profile may also make the detection of the fundamental line more difficult than at higher harmonics, such as Vela X-1 and A0535+26. Moreover, the ratio of the line energies seems to be larger than 2 in Vela X-1 (Kreykenbohm et al. 1999). As the number of the superposition of the original lines increases in our calculations, the ratio of the line energies tends to become larger than 2 and a broader, shallower, and more complex fundamental feature can be produced. These properties of the line profile may lead to the absence of the fundamental line in observation. These properties are also compatible with the harmonic ratio of the line energies calculated by our model, as discussed below.

Tsygankov et al. (2006) reported that there is a possibility that the ratio of the line energies of the second harmonic to the fundamental slightly decreases with decreasing luminosity in V0332+53 observed with *INTEGRAL* and *RXTE*, although they cannot make a final conclusion about it due to lower statistics of the data at the high energies. Nishimura (2003) pointed out that the ratio of the second to the fundamental of the peak energy in the cyclotron absorption line becomes more than the harmonic ratio in the extended line-forming region threaded by a dipole field. This is because the cross section of the second harmonic is much smaller than that of the fundamental and there are some differences between the second harmonic and the fundamental in the line formation mechanism. Figure 11 displays the ratios of the second harmonic to the fundamental in the peak energies of the cyclotron absorption lines as a function of the shock height in the cylinder geometry with one angle  $\mu = 0.57$  in our model. As the

shock height, i.e., the luminosity, increases, the harmonic ratio tends to increase as shown in Figure 11, except for the shock height  $2.8 \times 10^5$  and  $3.3 \times 10^5$  cm (the cyclotron energy ranges 8–16 and 9–16 keV). The original individual second harmonic absorption feature becomes even shallower than the original individual fundamental with the height of the accretion column with the density gradient decreasing with the height as shown in Figure 3, since the cross section of the second harmonic is much smaller than that at the fundamental. However, the main reason is that the third harmonic absorption line gets to mix with the second harmonic absorption line as the number of the superposition of the lines, so that the peak energy of the second harmonic absorption line tends to shift toward higher energies. Consequently, the peak energy of the resulting second harmonic absorption feature indicates the magnetic field strength of relatively higher energy than at the fundamental. Thus, it tends to become larger than twice that of the fundamental as the number of the superposition of the lines increases as shown in Figure 11: the harmonic ratio would therefore tend to increase with an extension of the line-forming region, although it decreases in the shock height  $2.8 \times 10^5$  and  $3.3 \times 10^5$  cm (the cyclotron energy ranges 8–16 and 9–16 keV). Thus, the harmonic line ratio varies with the extension of the line-forming region in our model.

### 5.2. Properties of Superposition of Cyclotron Line Features

We found that the superposition of some lines with different peak energies emerging from some individual domains changes deeper and narrower features of the original lines to shallower and broader. In a single cylinder or slab geometry, it is likely difficult to reproduce such a broader and shallower line required for accretion-powered pulsars, except for the 1–0 slab geometry with an assumed magnetic field increasing linearly with height, such as a nondipole field, or a distance from an emission region (Nishimura 2005). In contrast with Nishimura (2005), the superposition of some lines emerging from some individual line-forming regions, which possess a different magnetic field strength at each domain, naturally leads to the development of broader and shallower line features as observed commonly in accreting X-ray pulsars. The fundamental line also tends to have the asymmetric profile that is shallower toward lower energies due to the gradient in the density and the thermal Doppler broadening. This fundamental line profile seems to be in good agreement with one observed recently in V0332+53 with *RXTE* (Pottschmidt et al. 2005). This property of the fundamental line is also relatively compatible with a distribution of the magnetic field derived by model fits for the spectra of 4U 1538–52 by Bulik et al. (1992).

The shape of the fundamental line can be more complex due to the superposition of a large number of the original cyclotron features with the absorption-like feature and two emission-like wings formed in the cylinder geometry. These characteristics may introduce some difficulties into spectral fitting to the observed fundamental features. When the extent of the line-forming region is sufficiently small to neglect the influence of the variation of the magnetic field strength on the line profile, the line profile may be approximated by a simple Gaussian profile. On the other hand, the second and third harmonic lines are also broader and shallower keeping one clear absorption structure unlike the fundamental, while it has an asymmetric shape that is shallower toward lower energies like the fundamental. The higher harmonic lines are expected to be able to be fitted by a simpler line shape, such as a Gaussian profile, than in the fundamental lines.

The angle-dependent spectra are computed by accumulating the photon densities emerging from each of the individual single slabs for the same photon's propagation angle with respect to the

field,  $\theta$ , as shown in Figures 9 and 10. The resulting fundamental line feature becomes deeper as it gets broader with the increasing viewing angles,  $\mu$ , in the slab geometry as well as the cylinder. We therefore conclude that this trend can be independent of the geometry, slabs or cylinders, in the cyclotron line formed by the superposition of some lines (the width of the cyclotron energy range  $\delta E_{\text{cyc}} \geq 3$  keV). This feature is in agreement with one observed for all accreting pulsars with *RXTE* (Coburn et al. 2002), where deeper CRSFs are also broader at least for the fundamental features, even when scaled by the centroid energy.

## 6. CONCLUSIONS

The cyclotron lines were computed by the sum of the photon densities emerging from some individual line-forming regions with a different magnetic field strength, temperature, and density. This formation mechanism for the cyclotron lines naturally leads to the development of broader and shallower line profiles as observed commonly in accreting X-ray pulsars, while it is likely difficult to accomplish with the previous models employing a single slab or cylinder. Moreover, the shape of the asymmetric profile that is shallower toward lower energies can be reproduced by the gradient in the thermal Doppler broadening. In addition, the line features tend to become deeper as they get broader with the viewing angles  $\mu$ , regardless of the geometries, slabs or cylinders. The superposition of the cyclotron lines emerging from a large number of the line-forming regions tends to diminish the angle dependence of the depth and width of the cyclotron line on the geometry, while the shape of the line depends on the geometry. Our resulting spectra exhibit a tendency to decrease in the energy and depth of the cyclotron fundamental line with the increase of the luminosity, while its width first decreases and then

increases. These properties are roughly in agreement with the results changing with the time observed in Mowlavi et al. (2006). Moreover, the harmonic ratios of the second to the fundamental tend to decrease with the increase of the luminosity in our calculations. This trend is consistent with those observed in the transient X-ray pulsar V0332+53 with *INTEGRAL* and *RXTE*.

Our model also succeeded in reproducing generally the property of the variations on the energy, depth, and width in the fundamental feature as observed in V0332+53 and X0115+63. However, the total amount of the variation on the energy of the fundamental line in our model is less than that in the observations for X0115+63 and more than for V0332+53. This might be attributed to the use of a Gaussian modeling in determining the cyclotron resonance energies in the observational data, although the cyclotron features at the fundamental can be so complex that it is likely difficult to be fitted by a Gaussian profile as in model fits to the spectra of 4U 1538–52 and Vela X-1 (Bulik et al. 1992, 1995). This may also imply the possibility of the presence of an effective zone to the line formation in the accretion column. If so, we need to consider the effective area of the accretion column to the line formation. Although our model only approximates the real situation, our model exhibits that the superposition of a large number of cyclotron lines from different heights or locations of the line-forming region can naturally lead to several properties of the cyclotron lines, as well as broader, shallower, and more complex profiles as observed commonly in a number of accretion-powered X-ray pulsars.

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